

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

COLLEGE OF ENGINEERING

B.Sc. (Engineering), Supplementary Examination, August, 2016

First (1st) Year

COE/EE 152: BASIC ELECTRONICS

TIME: 2 HRS 30 MINS

PROGRAM OF STUDY:

INDEX NO:.....

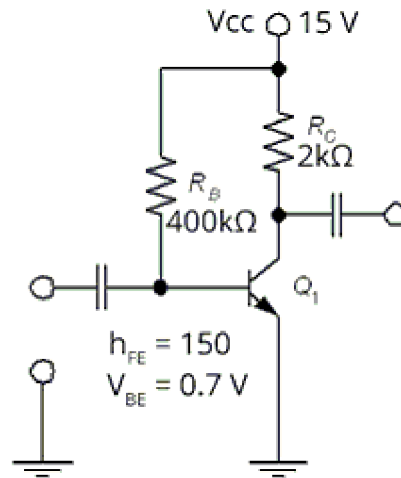
Instructions

- i. Candidates are to indicate their Index Number and Program of Study in the spaces provided.*
- ii. Candidates are to submit the question booklet together with all rough work if any.*
- iii. Circle and indicate, from the options lettered A to D, the most suitable answer to the question statements*
- iv. Shade the most suitable answer to the question statements on the Scannable Sheet provided*
- v. Each correct answer carries half ($\frac{1}{2}$) a mark*

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

- 1) In the saturation region, the base-emitter junction _____.
 A) and the base-collector junctions are both forward-biased
 B) is reversed-biased while the base-collector junction is forward-biased
 C) is forward-biased while the base-collector junction is reversed-biased
 D) and the base-collector junctions are both reverse-biased
 1) _____
- 2) A(n) _____ is added to the fixed-bias configuration to improve bias stability.
 A) base voltage
 B) collector resistor
 C) emitter resistor
 D) All of the above
 2) _____

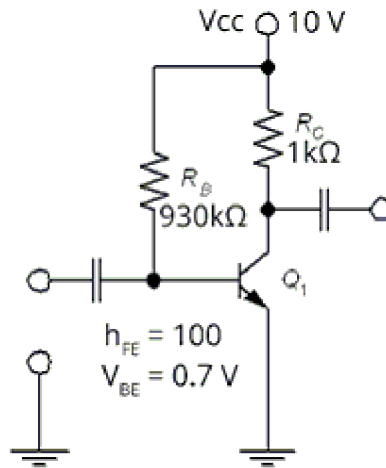
Use the fixed bias circuit below to solve questions 3 to 7



- 3) Determine the value of the base current, I_B
 A) 35.75mA
 B) 35.75 μ A
 C) 36.75mA
 D) 36.75 μ A
 3) _____
- 4) Determine the value of the collector current, I_C
 A) 5.125mA
 B) 5.125A
 C) 5.363mA
 D) 5.363A
 4) _____
- 5) Determine the value of the collector-emitter voltage, V_{CE}
 A) — 4.275V
 B) 4.275mV
 C) 42.75V
 D) 4.275V
 5) _____
- 6) Determine the value of the collector current at saturation, $I_{C(Sat)}$
 A) 7.5mA
 B) 7.5A
 C) 1.5 μ A
 D) 15A
 6) _____
- 7) Determine the value of the collector-emitter voltage at cut-off, $V_{CE(off)}$
 A) 1.5V
 B) 15V
 C) 7.5V
 D) 75V
 7) _____
- 8) Pentavalent atoms are often referred to as _____.
 A) donor atoms
 B) majority carriers
 C) acceptor atoms
 D) minority carriers
 8) _____

- 9) When voltage-divider bias is used, it is considered appropriate to use the approximate analysis to determine the bias condition when the resistance R_2 is _____ ($1+\beta$) R_E . 9) _____
- A) very much less than B) greater than
C) less than D) very much greater than

Use the fixed bias circuit below to solve questions 10 to 15

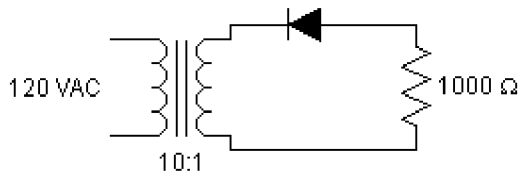


- 10) Determine the value of the base current, I_B 10) _____
A) 10mA B) 0.1mA C) 100mA D) 10μA
- 11) Determine the value of the collector current, I_C 11) _____
A) 1A B) 10A C) 1mA D) 10mA
- 12) Determine the value of the emitter current, I_E 12) _____
A) 1.01A B) 10.1A C) 1.01mA D) 10.1mA
- 13) Determine the value of the collector-emitter voltage, V_{CE} 13) _____
A) 9V B) 9.99mV C) 9.99V D) — 990V
- 14) Determine the value of the collector current at saturation, $I_{C(Sat)}$ 14) _____
A) 1mA B) 10A C) 10μA D) 10mA
- 15) Determine the value of the collector-emitter voltage at cut-off, $V_{CE(off)}$ 15) _____
A) 1V B) 10V C) 100V D) 10mV
- 16) What denotes the maximum voltage a diode can withstand when reverse-biased? 16) _____
A) Peak inverse voltage B) Peak forward-bias voltage
C) Reverse breakdown D) Reverse bias
- 17) When a BJT has its base-emitter junction forward biased and its collector-base junction reverse biased, it is biased in the _____. 17) _____
A) cutoff region B) saturation region
C) passive region D) active region

- 18) In the active region, the base-emitter junction _____. 18) _____
- A) and the base-collector junctions are both forward-biased
 - B) and the base-collector junctions are both reverse-biased
 - C) is reverse-biased while the base-collector junction is forward-biased
 - D) is forward-biased while the base-collector junction is reverse-biased

- 19) Why is design for a specific bias point desirable for most amplifiers? 19) _____
- A) To meet manufacturer suggested operating point.
 - B) It allows optimum dc operation of the circuit.
 - C) It allows optimum ac operation of the circuit.
 - D) All of the above

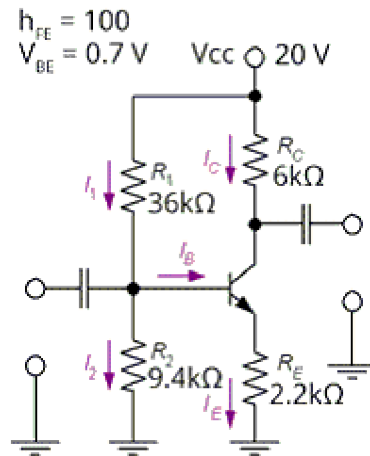
- 20) Calculate the peak current that will flow through this circuit, assuming an ideal diode. 20) _____



- A) 16.97 mA during the positive half cycle
 - B) 12 mA during the negative half cycle
 - C) 12 mA during the positive half cycle
 - D) 16.97 mA during the negative half cycle
- 21) The reverse saturation current of a diode will just about _____ for every 10°C rise in the diode temperature. 21) _____
- A) decrease proportionately with temperature
 - B) increase proportionately with temperature
 - C) double
 - D) half
- 22) For basic operation of a transistor the base-emitter junction is _____ biased. 22) _____
- A) reverse- B) forward- C) semi- D) not
- 23) A BJT has measured dc current values of $I_B = 0.1 \text{ mA}$ and $I_C = 8.0 \text{ mA}$. When I_B is varied by 100 μA , I_C changes by 10 mA. What is the value of the β_{dc} for this device? 23) _____
- A) 800 B) 100 C) 10 D) 80

- 24) The DC or the static resistance of the diode is given by _____. 24) _____
- A) $R_D = \frac{V_{D1} - V_{D2}}{I_{D1} - I_{D2}}$ B) $R_D = \frac{V_D}{I_D}$
- C) $R_D = \frac{V V_D}{V I_D}$ D) All of the above can be used.

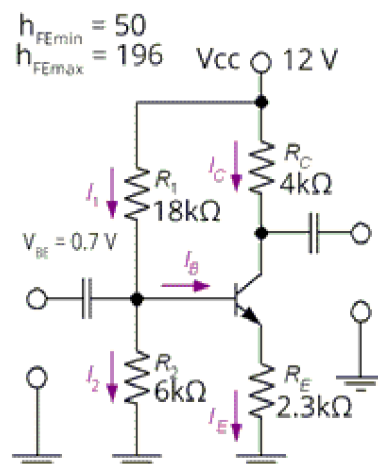
Use the voltage divider bias circuit below to solve questions 25 to 33. Assume that $h_{FE} R_E > 10 R_2$



- 25) Determine the value of the base voltage, V_B 25) _____
- A) 4.40V B) 2.13V C) 4.14V D) 41.4V
- 26) Determine the value of the emitter voltage, V_E 26) _____
- A) 1.34V B) 3.44V C) 1.43V D) 3.70V
- 27) Determine the value of the emitter current, I_E 27) _____
- A) 0.59mA B) 0.65mA C) 1.56mA D) 0.61mA
- 28) Determine the value of the collector current at the Q-point, I_{CQ} 28) _____
- A) 0.59mA B) 0.65mA C) 1.56mA D) 0.61mA
- 29) Determine the value of the collector-emitter voltage at the Q-point, V_{CEQ} 29) _____
- A) 15.162V B) 10.616V C) 16.559V D) 7.175V
- 30) Determine the value of the current across the resistor R_2 , I_2 30) _____
- A) 468.09μA B) 226.59μA C) 440.53μA D) 440.53mA
- 31) Determine the value of the base current, I_B 31) _____
- A) 0A B) 440.53μA C) 4.4053μA D) 15.485μA
- 32) Determine the value of the collector current at saturation, $I_{C(sat)}$ 32) _____
- A) 3.333mA B) 9.091mA C) 2.439mA D) 1.549mA
- 33) Determine the value of the collector-emitter voltage at cut-off, $V_{CE(off)}$ 33) _____
- A) 0V B) 20V C) 10V D) 9.09V

- 34) A given transistor has ratings of maximum collector current equal to 200 mA and a beta that varies between 150 and 200. What is the maximum allowable value of base current for the device? 34) _____
 A) 1 mA B) 4 mA
 C) 1.33 mA D) None of the above
- 35) Which amplifier configuration is the least often used? 35) _____
 A) Common Base B) Common Collector
 C) Darlington Pair D) Common Emitter
- 36) When a Zener diode circuit is used to stabilize the output voltage given a fixed load resistor and a variable input voltage, the input voltage must be _____. 36) _____
 A) large enough to turn off the Zener diode B) small enough to turn on the Zener diode
 C) small enough to turn off the Zener diode D) large enough to turn on the Zener diode
- 37) Transistor circuits that are quite stable and relatively insensitive to temperature variations have _____. 37) _____
 A) relative high supply voltages B) small betas
 C) large betas D) low supply voltages
- 38) If V_{CE} is 13 V and V_{CC} is 15 V, this transistor is 38) _____
 A) properly biased B) approaching cutoff
 C) a pnp transistor D) approaching saturation
- 39) Which transistor amplifier configuration is the most commonly used? 39) _____
 A) common-collector
 B) common-emitter
 C) common-base
 D) None of these are used more often than the others.
- 40) An open diode in a bridge rectifier circuit will produce what type of waveform? 40) _____
 A) Half-wave with decreased ripple B) Half-wave with increased ripple
 C) Full-wave with increased ripple D) Full-wave with decreased ripple

Use the voltage divider bias circuit below to solve questions 41 to 50. Assume that $h_{FE}R_E > 10R_2$

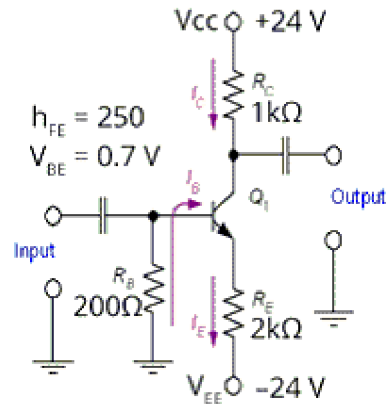


- 41) Determine the value of $h_{FE(ave)}$ 41) _____
 A) 196 B) 123 C) 11 D) 99
- 42) Determine the value of the base voltage, V_B 42) _____
 A) 3V B) 2V C) 1V D) 0.5V
- 43) Determine the value of the emitter voltage, V_E 43) _____
 A) 3.7V B) 1.3V C) 0.3V D) 2.3V
- 44) Determine the value of the emitter current, I_E 44) _____
 A) 1.61mA B) 1.0mA C) 0.13mA D) 10mA
- 45) Determine the value of the collector current at the Q-point, I_{CQ} 45) _____
 A) 1.61mA B) 1.0mA C) 0.13mA D) 10mA
- 46) Determine the value of the collector-emitter voltage at the Q-point, V_{CEQ} 46) _____
 A) 5.7V B) 1.92V C) 11.81V D) 3.7V
- 47) Determine the value of the current across the resistor R_2 , I_2 47) _____
 A) 333.33 μ A B) 166.66 μ A C) 450 μ A D) 500 μ A
- 48) Determine the value of the base current, I_B 48) _____
 A) 0A B) 19.608 μ A C) 10 μ A D) 5.102 μ A
- 49) Determine the value of the collector current at saturation, $I_{C(sat)}$ 49) _____
 A) 1.905mA B) 3mA C) 5.217mA D) 6.66mA
- 50) Determine the value of the collector-emitter voltage at cut-off, $V_{CE(off)}$ 50) _____
 A) 0V B) 10V C) 12V D) 20V
- 51) Which of the following biasing combinations is not normally associated with one of the three transistor operating regions'? 51) _____
 A) E-B junction = forward, C-B junction = reverse
 B) E-B junction = reverse, C-B junction = reverse
 C) E-B junction = reverse, C-B junction = forward
 D) All of the above
- 52) When a $p-n$ junction is reverse-biased, the depletion layer is _____ and the device acts as a near-perfect _____. 52) _____
 A) narrowed; insulator B) widened; insulator
 C) narrowed; conductor D) widened; conductor

- 53) The term _____ is applied to any material that supports a generous flow of charge when a voltage source of limited magnitude is applied across its terminals.
- A) semiconductor B) dielectric C) insulator D) conductor

53) _____

Use the emitter bias circuit below to solve questions 54 to 58



- 54) Determine the value of the base current, I_B
- A) $47.789\mu\text{A}$ B) $46.396\mu\text{A}$ C) 0A D) $95.579\mu\text{A}$
- 55) Determine the value of the collector current at the Q-point, I_{CQ}
- A) 1.61mA B) 11.6mA C) 23.9mA D) 11.9mA
- 56) Determine the value of the collector-emitter voltage at the Q-point, V_{CEQ}
- A) 4.8V B) 71.7V C) 13.2V D) 10.8V
- 57) Determine the value of the collector current at saturation, $I_{C(\text{Sat})}$
- A) 8mA B) 16mA C) 24mA D) 48mA
- 58) Determine the value of the collector-emitter voltage at cut-off, $V_{CE(\text{off})}$
- A) 0V B) 24V C) 12V D) 48V

54) _____

55) _____

56) _____

57) _____

58) _____

- 59) In the design of an emitter-bias stabilized circuit engineering, judgment must be used because the _____.

59) _____

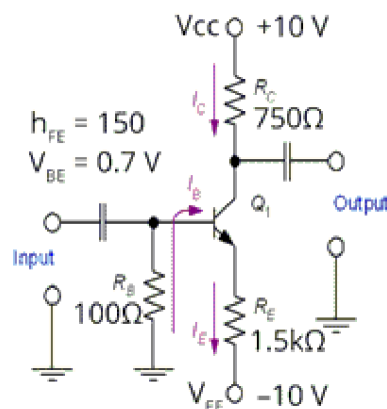
- A) relative voltage levels have not been defined
B) emitter resistor is usually unknown
C) collector resistor is usually unknown
D) All of the above

- 60) There are transistors that are called switching transistors because _____.
- A) of the speed at which they can be changed from on to off
B) of the voltage they can transfer from input to output
C) of the power they can transfer from input to output
D) they have a built in switch

60) _____

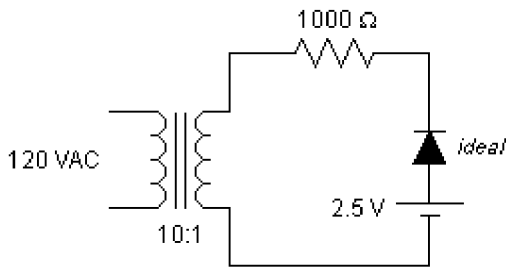
- 61) The majority carriers in the p region of a diode _____
 A) are much fewer in number than the majority carriers in the n region
 B) are electrons
 C) are much less in number than the majority carriers in the n region
 D) are holes
- 62) Doping is used to _____.
 A) increase the conductivity of an intrinsic semiconductor
 B) increase the insulative quality of an intrinsic semiconductor
 C) stabilize the conductivity of an intrinsic semiconductor
 D) decrease the conductivity of an intrinsic semiconductor
- 63) The term bipolar means:
 A) the use of both holes and electrons as carriers
 B) electrons make up the majority carriers
 C) polarity is unimportant
 D) diodes only
- 64) Voltage-divider bias stability is _____.
 A) dependent on the collector resistor
 B) independent of beta
 C) dependent on alpha
 D) dependent of beta
- 65) Collector-feedback bias _____.
 A) is not totally independent of beta
 B) provides an improved level of stability over fixed-bias
 C) provides a feedback path from collector to base
 D) All of the above

Use the emitter bias circuit below to solve questions 66 to 70



- 66) Determine the value of the base current, I_B _____
 A) $93\mu\text{A}$ B) $44.150\mu\text{A}$ C) 0A D) $41.04\mu\text{A}$
- 67) Determine the value of the collector current at the Q-point, I_{CQ} _____
 A) 6.156mA B) 13.95mA C) 6.623mA D) 4.415mA

- 68) Determine the value of the collector-emitter voltage at the Q-point, V_{CEQ} 68) _____
 A) 3.851V B) 6.149V C) 16.149V D) 20V
- 69) Determine the value of the collector current at saturation, $I_{C(sat)}$ 69) _____
 A) 8.89mA B) 13.33mA C) 4.44mA D) 44.44mA
- 70) Determine the value of the collector-emitter voltage at cut-off, $V_{CE(off)}$ 70) _____
 A) 10V B) 0V C) 20V D) 40V
- 71) A typical Zener diode regulator circuit uses a _____. 71) _____
 A) resistor in parallel with the load
 B) dropping resistor in series with the load
 C) Zener diode in parallel with the series resistor
 D) Zener diode in series with the load
- 72) When a BJT has its base-emitter junction reverse biased and its collector-base junction reverse biased, it is in the _____. 72) _____
 A) saturation region B) cutoff region
 C) active region D) passive region
- 73) When a BJT is operating in the saturation region the voltage drop from the collector to the emitter V_{CE} is approximately equal to _____. 73) _____
 A) the collector supply voltage
 B) the collector current times the collector resistor
 C) zero (about 0.3 Volts)
 D) the emitter voltage
- 74) The maximum reverse bias potential that can be applied to a Zener diode before it enters the Zener region is called the _____. 74) _____
 A) threshold voltage B) PIV
 C) depletion voltage D) barrier voltage
- 75) For this clipping circuit, what is the maximum output voltage when the diode is not conducting? 75) _____



- A) + 19.47 V B) - 16.97 V C) + 16.97 V D) + 2.5 V
- 76) When a $p-n$ junction is reverse-biased, its junction resistance is _____. 76) _____
 A) low
 B) high
 C) constantly changing
 D) determined by the components that are external to the device

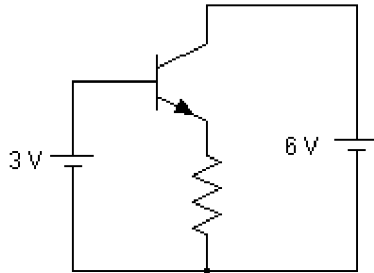
77) What can be said regarding the energy level of an electron?

77) _____

- A) Electrons exhibit energy characteristics only in the presence of protons
- B) Electrons in the shell closest to the nucleus of an atom have the highest energy level
- C) Electrons in the outermost shell have the highest energy level
- D) All electrons have the same energy level

78) Which of the following is true for this BJT circuit?

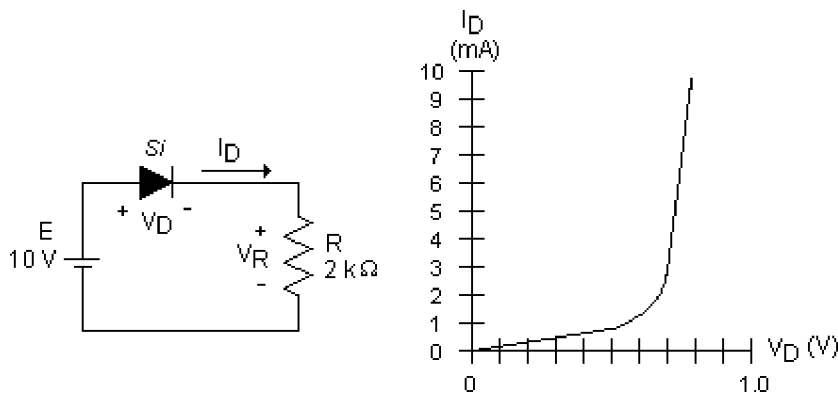
78) _____



- A) The base-emitter and collector-base junctions are both reverse-biased.
- B) The base-emitter and collector-base junctions are both forward-biased.
- C) The base-emitter junction is reverse-biased and the collector-base junction is forward-biased.
- D) The base-emitter junction is forward-biased and the collector-base junction is reversed-biased.

79) For this series diode configuration, use the diode characteristic to estimate the value of V_R .

79) _____



- A) 92 mV
- B) 9.2 V
- C) 0.92 V
- D) 10 V

80) When a BJT has its base-emitter junction forward biased and its collector-base junction also forward biased, it is in the _____.

80) _____

- A) active region
- B) passive region
- C) cut-off region
- D) saturation region

81) Two of the factors associated with bias stability are _____.

81) _____

- A) the β and the junction temperature
- B) age and amount of use
- C) voltage and current
- D) None of the above

82) A full-wave center-tapped rectifier has a secondary maximum voltage of $20 V_m$ and a $4.7 k\Omega$ load resistance. What is the dc load current for the circuit?

82) _____

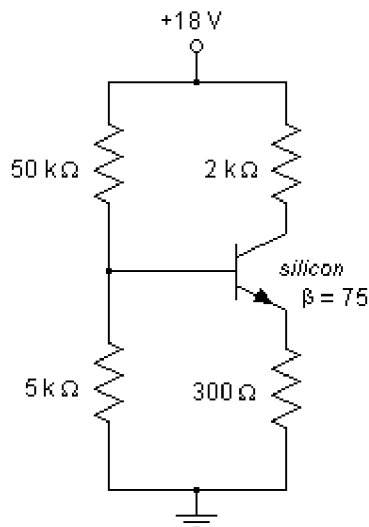
- A) 1.4 mA
- B) 1.26 mA
- C) 2.61 mA
- D) 629.8 mA

83) In a p -type material, the minority carriers are _____. 83) _____
 A) conduction-band holes B) valence-band holes
 C) valence-band electrons D) conduction-band electrons

84) A BJT has measured dc current values of $I_B = 1 \text{ mA}$ and $I_C = 80 \text{ mA}$. When I_B is varied by $100 \mu\text{A}$, I_C changes by 10 mA . What is the value of β_{ac} for the device? 84) _____
 A) 100 B) 80 C) 10 D) 800

85) What is the purpose of a current-limiting resistor in a diode circuit? 85) _____
 A) A current-limiting resistor is only necessary if it is reverse biased
 B) The resistor prevents the diode from becoming reverse-biased
 C) A properly biased diode is like a short circuit, therefore a resistor provides a voltage drop
 D) The resistor will increase the PIV rating of the diode

86) Calculate the base current for this voltage-divider bias circuit. 86) _____



A) $76.8 \mu\text{A}$ B) $596.55 \mu\text{A}$ C) $34.37 \mu\text{A}$ D) $233.78 \mu\text{A}$

87) $V_{CC} = 12 \text{ volts}$, $R_C = 1\text{k}$, determine the load line I_C and V_{CE} intercepts 87) _____
 A) $I_C = .012\text{mA}$ and $V_{CE} = 1.2 \text{ volts}$ B) $I_C = I_B = 12\text{mA}$ and $V_{CE} = 12\text{m volts}$
 C) $I_C = 12\text{mA}$ and $V_{CE} = 12 \text{ volts}$ D) $I_C = 1.2\text{mA}$ and $V_{CE} = 12\text{m volts}$

88) A p - n junction is forward biased when _____. 88) _____
 A) the applied potential causes the n -type material to be more negative than the p -type material
 B) the applied potential causes the n -type material to be more positive than the p -type material
 C) both materials are at the same potential
 D) None of these

89) When in its "on" state, the voltage across an ideal Zener diode, V_Z _____. 89) _____
 A) gets larger with an increase in applied voltage
 B) gets smaller with an increase in applied voltage
 C) increases sharply with a decrease in applied voltage
 D) None of these

- 90) For a given BJT, $\beta = 400$. What is the value of α for the device? 90) _____
 A) 1.00 B) 1.0025 C) 0.002 D) 0.9975
- 91) When a BJT is biased in the active region, its base-emitter junction is _____ -biased and its collector-base junction is _____ -biased. 91) _____
 A) reverse; forward B) forward; reverse
 C) reverse; reverse D) forward; forward
- 92) A Zener diode is designed to operate in the _____ region of its characteristic curve. 92) _____
 A) reverse breakdown B) zero voltage
 C) reverse bias D) forward operating
- 93) When trivalent elements are used in doping, the resulting material is called _____ material and has an excess of _____. 93) _____
 A) p -type; conduction-band electrons B) n -type; valence-band holes
 C) p -type; valence-band holes D) n -type; conduction-band electrons
- 94) When recombination occurs, an electron 94) _____
 A) loses energy B) drops into the valence band
 C) drops into a hole D) all of the above
- 95) Collector feedback bias works on what principle? 95) _____
 A) falling collector currents tend to attenuate base current
 B) rising collector currents tend to attenuate base current
 C) base current will always remain constant
 D) positive feedback adds to collector current
- 96) The emitter-follower configuration has _____. 96) _____
 A) a 180° phase shift
 B) the emitter connected to dc ground potential
 C) an output voltage slightly greater than the input voltage
 D) None of the above
- 97) The typical range of the ac resistance of a diode in the active region is _____. 97) _____
 A) $100\ \Omega$ to $500\ \Omega$ B) $1\ \Omega$ to $10\ \Omega$ C) $1\ \Omega$ to $100\ \Omega$ D) $50\ \Omega$ to $100\ \Omega$
- 98) In a Common-Emitter amplifier 98) _____
 A) the emitter is always above ground potential
 B) the emitter is the reference for the input signal
 C) the output is taken from the collector
 D) the output signal is taken from the emitter
- 99) An npn BJT has the following values measured: $V_C = 8.3\text{ V}$, $V_E = 1.1\text{ V}$ and $V_{CC} = 18\text{ V}$. The value of V_{CE} must be 99) _____
 A) Cannot be determined, it depends on the value of β .
 B) 8.6 V
 C) 9.4 V
 D) 27.4 V

- 100) When pentavalent elements are used in doping, the resulting material is called _____ material and has an excess of _____.
 A) *p*-type; conduction-band electrons
 B) *p*-type; valence-band holes
 C) *n*-type; valence-band holes
 D) *n*-type; conduction-band electrons
- 101) When a BJT is biased in the cut-off region, its base-emitter junction is _____-biased and its collector-base junction is _____-biased.
 A) forward; reverse
 B) forward; forward
 C) reverse; reverse
 D) reverse; forward
- 102) A given BJT has an emitter current of 15 mA and a collector current of 14.95 mA. What is the exact value of β ?
 A) 300
 B) 250
 C) 1.003
 D) 299
- 103) If the base of a transistor is open, the voltage at the collector will be
 A) it depends on the beta of the transistor
 B) V_{CC}
 C) $V_{CC}/2$
 D) V_B
- 104) Increasing the temperature of a forward-biased diode _____.
 A) causes forward current to increase
 B) causes forward current to decrease
 C) has no significant effect on the forward current
 D) None of these
- 105) When a BJT has its base-emitter junction reverse biased and its base-collector junction forward biased, it is biased in the _____.
 A) saturation region
 B) cutoff region
 C) passive region
 D) active region
- 106) As the device temperature increases, semiconductor materials tend to have _____.
 A) an increasing number of free electrons
 B) lower conduction levels
 C) relatively unchanged conduction levels
 D) a decreasing number of free electrons
- 107) In a small-signal transistor, the typical range of the parameter β is _____.
 A) large and in the range of about 50 to 400
 B) between 0 and 100
 C) greater than 100
 D) almost equal to 100 but always less than 100 (90 to 100)
- 108) The *common-emitter, forward-current, amplification factor* is better known as _____.
 A) $dc \beta$
 B) $dc \alpha$
 C) $ac \alpha$
 D) $ac \beta$
- 109) The Zener diode must be operated such that _____.
 A) $I_Z \times V_Z = P_Z$
 B) the applied voltage is greater than V_Z
 C) P_Z is less than the specified P_{Zmax}
 D) All of these

- 110) A p - n junction is reverse biased when _____. 110) _____
 A) the current flow across the junction is based on minority carrier transfer
 B) the applied potential causes the n -type material to be more positive than the p -type material
 C) the applied potential causes the p -type material to be more negative than the n -type material
 D) All of the above
- 111) Which best describes a conducting forward-biased diode? 111) _____
 A) Negative cathode, positive anode $V_F = 0.7$ volts
 B) Positive cathode, negative anode $V_F = .2$ volts
 C) Negative cathode, positive anode $V_F = .2$ volts
 D) Positive cathode, negative anode $V_F = 10$ volts
- 112) If one silicon diode and one germanium diode are connected in series, the voltage drop across the combination of the two diodes will be equal to _____. 112) _____
 A) the forward drop equal to that of the germanium diode
 B) the forward drop equal to that of the silicon diode
 C) the forward drop equal to that of the difference of the voltage drops across the two diodes
 D) the forward drop equal to that of the sum of the voltage drops across the two diodes
- 113) As semiconductor devices have become _____ one of the primary purposes of the container is simply to provide a means for physical handling. 113) _____
 A) miniaturized B) more powerful C) widely used D) larger
- 114) When the Zener regulator is used to stabilize the output voltage, given a fixed input voltage and a variable load resistance, a load resistance that is too small results in _____. 114) _____
 A) V_Z being equal to V_{in} B) V_L being greater than V_Z
 C) V_L being less than V_Z D) V_L being equal to V_Z
- 115) The diode electrode with p -type material is called the _____. 115) _____
 A) cathode B) Zener region
 C) depletion region D) anode
- 116) A positive full-wave center-tapped rectifier has a secondary voltage of $20 V_m$. The peak load voltage for the circuit is _____ if the diode drop is included. 116) _____
 A) $9.3 V_p$ B) $20 V_p$ C) $10 V_p$ D) $19.3 V_p$
- 117) In an n -type material, the majority carriers are _____. 117) _____
 A) valence-band holes B) valence-band electron
 C) conduction-band holes D) conduction-band electrons
- 118) An npn transistor is biased using collector feedback. $R_C = 2.2 \text{ k}\Omega$ and $R_B = 220 \text{ k}\Omega$. $V_{CC} = 25 \text{ V}$ and $\beta = 250$. Solve for the value of base current. 118) _____
 A) $31.6 \mu\text{A}$ B) 7.9 mA C) $214 \mu\text{A}$ D) $3.16 \mu\text{A}$
- 119) A BJT has measured dc current values of $I_B = 1 \text{ mA}$ and $I_C = 80 \text{ mA}$. When I_B is varied by $100 \mu\text{A}$, I_C changes by 10 mA . What is the value of β_{dc} for the device? 119) _____
 A) 80 B) 100 C) 800 D) 10

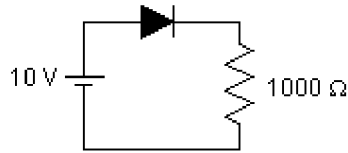
120) To design a transistor circuit for maximum stability, one must consider _____.

120) _____

- A) the transistor's beta stability factor
- B) the collector leakage current stability factor
- C) the base-emitter junction voltage stability factor
- D) All of the above

121) For this circuit, determine the load-line intersection with the two axis.

121) _____



- A) $V_D = 1\text{ V}$ and $I_D = 1\text{ mA}$
- B) $V_D = 10\text{ V}$ and $I_D = 10\text{ mA}$
- C) $V_D = 10\text{ V}$ and $I_D = 1\text{ mA}$
- D) $V_D = 1\text{ V}$ and $I_D = 10\text{ mA}$

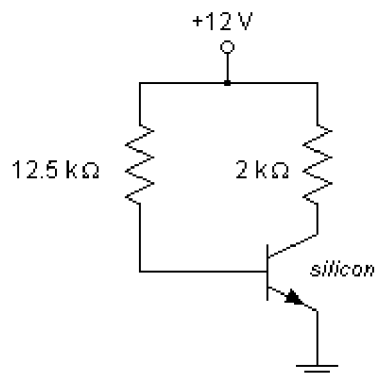
122) If a zener diode is connected to the base of a transistor, chances are it is a _____.

122) _____

- A) beta-stabilized circuit
- B) current mirror
- C) constant-current source
- D) gain-stabilized amplifier

123) Calculate the maximum collector current for this circuit.

123) _____



- A) 0.904 mA
- B) 0.056 mA
- C) 6.0 mA
- D) 0.96 mA

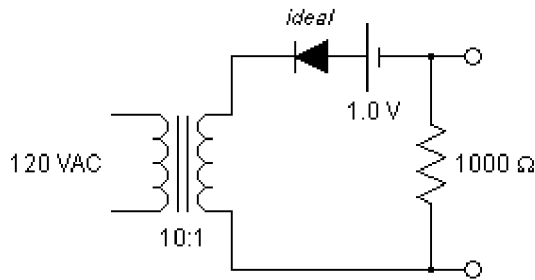
124) The ideal diode symbol has an arrow that points in the direction of _____.

124) _____

- A) the leakage current flow
- B) the forward current flow
- C) positive terminal under forward bias
- D) All of the above

125) For this clipping circuit, what is the minimum output voltage when the diode is conducting?

125) _____



A) + 16.97 V

B) - 1.0 V

C) - 17.97 V

D) - 16.97 V

126) The Zener diode is on if the applied voltage, V , is _____.

126) _____

A) $V < V_Z / 2$

B) $V \geq V_Z$

C) $V > 2V_Z$

D) $V < V_Z$

127) When a transistor is in saturation, the total collector current is limited by _____.

127) _____

A) collector supply, collector-to-emitter voltage, and the total collector circuit resistance

B) collector-to-emitter and collector supply voltage

C) the transistor

D) collector supply voltage and the total resistance in the collector and emitter circuits

128) The diode electrode with n -type material is called the _____.

128) _____

A) Zener region

B) anode

C) depletion region

D) cathode

129) The term *quiescent* means _____.

129) _____

A) active

B) inactive

C) at rest

D) midpoint-biased

130) The change in β and V_{CE} that can occur when the temperature changes is known as _____.

130) _____

A) Q-point movement

B) midpoint bias

C) midpoint movement

D) output movement

131) Variation in h_{fe} is influenced by _____.

131) _____

A) bias type and device size

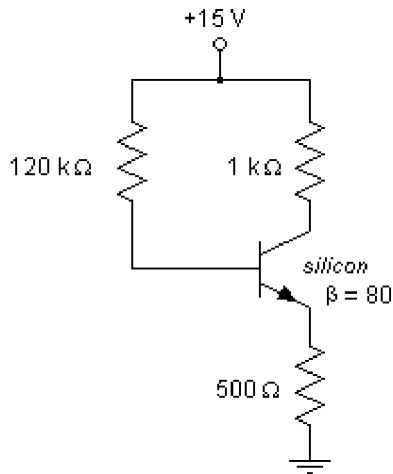
B) junction temperature and collector current

C) temperature and base current

D) device size and base current

132) Calculate the base current for this emitter-stabilized bias circuit.

132) _____



- A) 89.1 μ A
C) 0.119 mA

- B) 89.0 mA
D) None of the above

133) A given BJT has an alpha of 0.9985 and a collector current of 15 mA. What is the value of base current?

133) _____

- A) 15 mA
C) 15.15 mA

- B) 14.85 mA
D) None of the above

134) Which bias circuit is found chiefly in integrated circuit applications?

134) _____

- A) voltage divider bias
C) emitter bias

- B) collector feedback bias
D) base bias

135) When a BJT is biased in the saturation region, its base-emitter junction is _____-biased and its collector-base junction is _____-biased.

135) _____

- A) forward; reverse
C) forward; forward

- B) reverse; forward
D) reverse; reverse

136) A bridge rectifier has values of $V_M = 177$ V, turns ratio = 5 : 1, and $R_L = 500 \Omega$. What is the dc output voltage?

136) _____

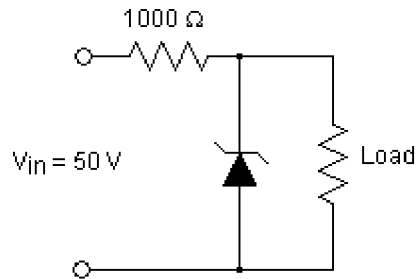
- A) 9.91 V

- B) 21.62 V

- C) 6.88 V

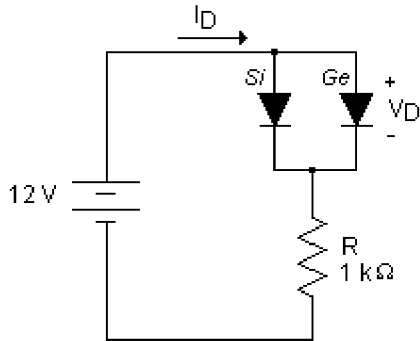
- D) 3.75 V

- 137) What are the minimum and maximum values of current flowing in the variable load resistor while the diode is operating in the Zener region? The zener voltage is 10 V. 137) _____



- A) 8 mA and 35 mA
 B) 8 mA and 40 mA
 C) 12.5 mA and 40 mA
 D) Need to know the load resistance to determine the values.

- 138) The value of V_D in this circuit is _____. 138) _____



- A) 11.3 V B) 0.3 V C) 0.7 V D) 10.6 V

- 139) A half-wave rectifier with the diode arrow pointing away from the load has a DC output voltage of _____ for an AC input voltage of 20 V maximum. 139) _____
- A) -6.14 V B) 19.3 V C) 12.49 V D) -13.65 V

- 140) Why are bridge rectifiers preferred over full-wave center-tapped rectifiers? 140) _____
- A) They provide higher dc output voltages.
 B) They do not require the use of a center-tapped transformer.
 C) They require a lower PIV rating.
 D) All the above

Answer Key

Testname: BASIC ELECTRONICS 2016_SUPPLEXAM

- 1) A
- 2) C
- 3) B
- 4) C
- 5) D
- 6) A
- 7) B
- 8) A
- 9) D
- 10) D
- 11) C
- 12) C
- 13) A
- 14) D
- 15) B
- 16) A
- 17) D
- 18) D
- 19) D
- 20) D
- 21) C
- 22) B
- 23) D
- 24) B
- 25) C
- 26) B
- 27) C
- 28) C
- 29) D
- 30) C
- 31) D
- 32) C
- 33) B
- 34) A
- 35) A
- 36) D
- 37) C
- 38) B
- 39) B
- 40) B
- 41) D
- 42) A
- 43) D
- 44) B
- 45) B
- 46) A
- 47) D
- 48) C
- 49) A
- 50) C

Answer Key

Testname: BASIC ELECTRONICS 2016_SUPPLEXAM

- 51) C
- 52) B
- 53) D
- 54) B
- 55) B
- 56) C
- 57) B
- 58) D
- 59) D
- 60) A
- 61) C
- 62) A
- 63) A
- 64) B
- 65) D
- 66) D
- 67) B
- 68) A
- 69) A
- 70) A
- 71) B
- 72) B
- 73) C
- 74) B
- 75) C
- 76) B
- 77) C
- 78) D
- 79) B
- 80) D
- 81) A
- 82) B
- 83) D
- 84) A
- 85) C
- 86) C
- 87) C
- 88) A
- 89) D
- 90) D
- 91) B
- 92) A
- 93) C
- 94) D
- 95) B
- 96) D
- 97) C
- 98) C
- 99) B
- 100) D

Answer Key

Testname: BASIC ELECTRONICS 2016_SUPPLEXAM

- 101) C
- 102) D
- 103) B
- 104) A
- 105) B
- 106) A
- 107) A
- 108) D
- 109) D
- 110) D
- 111) A
- 112) D
- 113) A
- 114) C
- 115) D
- 116) A
- 117) D
- 118) A
- 119) A
- 120) D
- 121) B
- 122) C
- 123) C
- 124) B
- 125) B
- 126) B
- 127) D
- 128) D
- 129) B
- 130) A
- 131) B
- 132) A
- 133) D
- 134) C
- 135) C
- 136) B
- 137) B
- 138) B
- 139) A
- 140) D